Probe-based target enrichment of SARS-CoV-2

1. Abstract

- Viral RNA library prep using **SMARTer Stranded Total RNA-Seq Kit v2 -Pico Input Mammalian** followed by a probe-based bait capture (SeqCap, Roche and xGen, IDT) to generate tagged enriched viral libraries from total RNA that retain directionality in the library.
- The **SMARTer Stranded** kit uses random primers, tailed with Illumina Read1 sequence, to start reverse transcription, and a Template Switching Oligo (TSO) to add Read2 sequence at the 3' of the synthesized cDNA. The 1st stranded cDNA is then amplified using Indexed primers to generate in one step 2nd strand cDNA and complete tagged Illumina libraries.
- Pooled libraries undergo target enrichment using custom virus-specific biotinylated probes to capture cDNA derived from viral RNA present in the sample.
- This method has been optimized for large scale viral sequencing projects, and up to 2 x 96-well RNA plates can be prepped in parallel.
- The protocol follows "**Option 2 (without fragmentation)"** workflow of the kit User Manual for library preparation, with ¼ of the recommended reaction volume for denaturation and cDNA synthesis, and ½ for PCR, all steps done in 384-well plates.

  NOTE: *the kit contains a Ribodepletion PCR-based module not used in this SOP, so PCR reagents are in excess and a higher volume for PCR can be used safely without depleting the kit of reagents unevenly.*
- The absence of an RNA fragmentation step coupled with more stringent cleanups (at 0.68x) generates libraries with longer insert size than previous SOPs, therefor this protocol is only to be used on RNA samples of high quality, and unnecessary freeze-thaw of the RNA **MUST** be avoided. As the RNA extracted from plasma is below the detection level for QC, if in doubt about the quality of the RNA provided, proceed with library preparation and pooling, but check with PMs before cleanup of the pool (step 2.8.2) as the 0.68X beads ratio recommended in this SOP would remove most of the library if the starting RNA is of lower quality.
- To maximize RNA input and improve sensitivity of the assay, an RNA concentration step at the start of the procedure has been added, though not necessary to generate libraries: the customer should advice if he/she is happy for the all RNA provided to be used for the prep.

1.1. Timeframe

The protocol can be completed in **2-3 days**, depending on the length of the hybridization time: if the capture is set up overnight on Day1, it can be completed, including the final mpx QC'd and submission, on Day2.

If there is not enough time to start the capture on Day1, set up the capture early on Day2 and hybridize for 4h: mind that once the hybridization is stopped, all following steps prior to LM-PCR must be completed!

NOTE*: the time to dry with the SpeedVac the libraries pooled on Day1 is variable and dependent on the volume of the pool, setting up an O/N capture could take >>1h*

Day1: Library prep with pooling, cleanup and QC; +/- set up for O/N capture

Day2: Capture; 10nM, +/- qPCR

	Step	**Estimated time**
Day1	*Preparation*	*30 minutes*
	RNA concentration	30 minutes
	cDNA synthesis setup	10 minutes
	cDNA 1st strand synthesis	*1h 40 minutes*
	PCR setup	15 minutes
	PCR	*35 minutes*
	Pooling by volume	10 minutes
	Pool cleanup	30 minutes
	QC	15 minutes
Optional Day1 or Day2	*SpeedVac drying of pool*	*20 minutes – 1.5 hours*
	Capture setup	20 minutes
Day2	*Probe Hybridization*	*4 hours (or overnight)*
	Preparation of streptavidin beads and wash buffers	15 minutes
	Hybridization to streptavidin beads	*45 mins*
	Streptavidin washes and PCR setup	20 mins
	Post-Capture LM-PCR	*35 mins*
	Capture cleanup	30 minutes
	QC and 10nM	15 mins
	Real-time quantitative PCR	30 minutes

2. Procedures

2.1. Material

2.1.1. Kit Reagents

Takara-Clontech, SMARTer Stranded Total RNA-Seq Kit v2 – Pico Input Mammalian

- SMART TSO Mix v2 (Cat#: ST1250)
- SMART Pico Oligos Mix v2 (Cat#: ST1262)
- 5X First-Strand Buffer (Cat#: ST1266)
- SMARTScribe RT (100 U/µl) (Cat#: ST1270)
- RNase Inhibitor (40 U/µl) (Cat#: ST1272)
- SeqAmp DNA Polymerase (Cat#: ST1280)
- SeqAmp CB PCR Buffer (2X) (Cat#: ST1282)

SeqCap Hybridization and Wash Kit (Cat#: 5634253001)

- 2X Stringent Wash Buffer
- 10X SC Wash Buffer I
- 10X SC Wash Buffer II
- 10X SC Wash Buffer III
- 2x Hybridization Buffer

xGen® Universal Blockers - TS Mix (Cat#: 1075475)
SeqCap EZ Accessory Kits v2 (Cat#: 07145594001)

- COT-1 Human DNA
- Hybridization Component A
- KAPA HiFi HotStart ReadyMix
- Post- LM-PCR Oligos 1 & 2

2.1.2. Non-kit Reagents

Gilson platemaster

- Gilson Platemaster - Pipette Filter Tips Sterilized (Cat#: DF30ST)
- Gilson Platemaster - Pipette Filter Tips Sterilized (Cat#: DF200ST)
- RNACleanXP (Cat#: A66514)
- AmpureXP (Cat#: A63881)
- IDT xGen® Lockdown Probes

2.2. Other required items

2.2.1. Equipment

- 4titude FrameStar® 96 Well Semi-Skirted PCR Plate (Cat#: 4ti-0900/C)
- 4titude FrameStar® 384 Well Clear PCR plate (Cat#: 4ti-0384/C)
- Invitrogen, DynaMag-2 Magnet (Cat#: 123-21D)
- 384 block Thermocycler
- DNA Vacuum Concentrator
- Heat block
- Single and multi-channel pipettes with tips, 2-1000µl

Qubit® 3.0 Fluorometer (Q33216)

- Qubit® dsDNA HS Assay Kit (Cat#: Q32854)

Tapestation 2200

- High Sensitivity D1000 Screen Tapes (Cat#: 5067-5584)
- High Sensitivity D5000 Screen Tapes (Cat#: 5067- 5593)
- High Sensitivity D1000 Reagent (Cat#: 5067-5585)
- High Sensitivity D5000 Reagent (Cat#: 5067- 5593)
- High Sensitivity D1000 Ladder 20µl (Cat#: 5067-5587)
- High Sensitivity D5000 Ladder 20µl (Cat#: 5067-5594)

2.3. Preparations

USE RNAse ZAP TO DECONTAMINATE ALL WORK SURFACES PRIOR TO PRACTICAL WORK, ONLY WORK IN A DEDICATED PRE-PCR AREA

2.3.1. Bring the RNAClean XP (A63987) to room temperature

2.3.2. Check that the metal chilling block is at -20 ˚C

2.3.3. Defrost reagents on ice ('SMARTer® Stranded Total RNA-Seq Kit v2 - Pico Input Mammalian stored at -20°C, SMART TSO Mix v2 stored at -80°C).

2.3.4. Prepare the following program on a 384 thermal cycler:

Program:
72°C ∞
72°C 3min
42°C ∞
42°C 90 min
70°C 10 min

4°C ∞

2.4. Concentration of the RNA using magnetic beads

2.4.1. Thaw the RNA samples on ice (up to 2 plates of 96 samples)

2.4.2. Add in RNAClean XP beads at 1.8 ratio to all RNA samples

NOTE: *precise measurement of the RNA volume of each sample is not crucial at this stage, as this "cleanup" has the only purpose of concentrating the RNA, and not removing unwanted oligos*

2.4.3. Incubate at RT for 5 minutes.

2.4.4. Place the plate on magnet: allow beads to separate for 8 minutes.

2.4.5. Remove supernatant and put aside in a new 96-well plate.

2.4.6. Add 200µl 80% ethanol to wash the beads, incubate for 30 seconds

2.4.7. Remove 80% EtOH

2.4.8. Repeat the ethanol wash (steps 2.4.6-2.4.7)

2.4.9. Briefly spin the plate and remove additional ethanol with fresh tips. Ensure any visible quantities of ethanol are removed.

2.4.10. If two 96-well plates are being prepared together, do both ethanol washes on the 1st plate before moving to the 2nd plate, so that the same PlateMaster tips can be used for both washes on a plate.

WARNING: Keep to time with washes on the 2nd plate to avoid over drying beads on the 1st!

2.4.11. Remove the plate(s) from magnet and resuspend beads in 3µl EB.

2.4.12. Leave the beads in EB for 5mins at room temperature

2.4.13. During the 5mins incubation, prepare a **384-well plate** containing 0.25µl of the SMART Pico Oligos Mix v2

2.5. RNA denaturation

2.5.1. Pre-heat the thermocycler to 72°C

2.5.2. Once the 5mins elution time is passed, place the RNA plate on the magnet and allow beads to separate for 2 minutes.

2.5.3. Transfer 2µl sample into the plate containing 0.25µl freshly-dispensed SMART Pico Oligos Mix v2

2.5.4. Mix with PlateMaster, quickly spin down the plate.

2.5.5. Final plate for denaturation should contain:

Reagent	Volume (µl)
SMART Pico Oligos Mix v2	0.25
RNA	2
Total	= 2.25µl / reaction

2.5.6. Place plate containing RNA and SMART Pico Oligos Mix v2 onto the pre-set thermocycler and incubate **72°C for 3 minutes.**

2.5.7. Immediately move samples to the chilling block and leave for 2 minutes

2.5.8. Proceed immediately to the next step

2.6. First-strand cDNA synthesis

2.6.1. Pre-heat thermal cycler to 42°C

2.6.2. Prepare First Strand reverse transcription mastermix by mixing the following reagents in the order shown (no. reactions + 10%):

Reagent	Volume per reaction (µl)
5X First-Strand Buffer	1
SMART TSO Mix v2	1.125
RNase Inhibitor	0.125
SMARTScribe Reverse Transcriptase	0.5
Total	2.75 + 2.25 sample = 5µl

2.6.3. Mix by pipetting and then spin down

2.6.4. Place in the pre-heated thermocycler for first strand RT incubation: **42°C for 90 minutes → 70°C for 10 minutes → 4°C hold.**

2.6.5. Leave samples at 4°C until next step.

SAFE STOPPING POINT! OVERNIGHT AT 4°C or STORE AT -20°C.

2.7. PCR amplification of cDNA and library generation

2.7.1. Prepare thermocycler in post-PCR for second strand amplification

2.7.2. Set up the PCR as follow:

	Component	Volume per reaction (µl)
Sample	*First-strand cDNA*	5
EACH	Indexed i7 primer (6.25µM)	1
	Indexed i5 primer (6.25µM)	1
Master Mix	Nuclease-free Water	5
	SeqAmp CB PCR Buffer (2X)	12.5
	SeqAmp DNA Polymerase	0.5
Total		5 samples + 2 primers + 18MM = 25 µl / reaction

2.7.3. Mix by pipetting, spin down the plate and start PCR:

Step	Temperature	Time	No. cycles
Initial Denaturation	94 °C	60 seconds	1
Denaturation	*98 °C*	*15 seconds*	
Annealing	*55 °C*	*15 seconds*	*12*

Extension	*68 °C*	*30 seconds*	
Final Extension	68 °C	2 minutes	1
Hold	10 °C	∞	

SAFE STOPPING POINT! OVERNIGHT AT 4°C or STORE AT -20°C.

2.8. Pooling, cleanup and QC

2.8.1. Pool in a 1.5ml LoBind tube equal volume of each sample to be captured together, according to the pooling strategy provided by the customer

NOTE: *library prepared with this protocol would yield on average 2-5 ng/µl: pool enough of each library for a final pool with >> 500ng (3 µl per sample for a 96-plex is usually plenty)*

2.8.2. Accurately measured with a pipette the volume of the pool: this is crucial for correct size-selection of the libraries with Ampure XP beads

2.8.3. Add 0.68x Ampure XP to each pool and perform a cleanup (SOP0378)

WARNING: the 0.68x ratio is to be used only on libraries from high quality RNA! If that is not the case, use 0.8x of Ampure XP instead

2.8.4. Mix thoroughly either by pipetting or vortexing.

2.8.5. Incubate for 5 minutes at room temp.

2.8.6. Transfer tubes to magnet and incubate for 8 minutes.

2.8.7. Remove and keep supernatant in a clean plate. Do not transfer any beads in the supernatant. If this is a risk, leave <5µl supernatant behind with the beads.

2.8.8. Wash beads with 200µl 80% ethanol. Leave for 30 seconds.

2.8.9. Remove and discard the supernatant. Repeat wash with ethanol.

2.8.10. Ensure removal of all ethanol.

2.8.11. Consider centrifuging plate briefly to collect any residual ethanol at the bottom of the well and pipette this off with a P10 or P20 pipette.

2.8.12. Air dry beads until the well looks dry and the beads are starting to crack.

2.8.13. Resuspend beads in a volume of EB equal to the starting pool volume or less.

NOTE: *eluting in less EB will speed the drying of the pool in the SpeedVac*

2.8.14. Mix by pipetting or vortexing. If vortexing, leave for 2 minutes before centrifuging.

2.8.15. Return tubes to magnet for 2 minutes.

2.8.16. Transfer the clean eluant to a fresh 1.5ml LoBind tube

2.8.17. Quantify the libraries pool using Qubit® dsDNA HS Assay Kit (Cat#: Q32854)

2.8.18. Check the size of the libraries pool on an Agilent Tapestation with a High Sensitivity D1000 Screen Tapes assay (Cat#: 5067-5584)

2.8.19. Bring forward 500ng from each pool for capture (Step 2.9) and store the remaining uncaptured pool.

2.9. Hybridize xGen Lockdown Probes to Target

2.9.1. Turn on a heat block that takes 1.5ml tubes, and let it equilibrate to 95°C

2.9.2. Defrost xGen® Universal Blockers (IDT), COT Human DNA (vial 1, Roche, SeqCap EZ Accessory Kit, stored at -20°C).

2.9.3. Bring the 2X Hybridization buffer (vial 5) and Hybridization Component A (vial 6) to room temperature (Roche, SeqCap EZ Hybridization and Wash Kit, stored at -20°C).

2.9.4. Remove the xGen Lockdown Probes (IDT) from -20°C freezer and defrost on ice

2.9.5. In a 1.5ml tube, add:

Component	Amount
Multiplex DNA Sample Library Pool	500ng
COT Human DNA	5µl
xGen® Universal Blockers - TS Mix	2µl

2.9.6. Dry down the contents of the tube (libraries + COT Human DNA + Blocking Oligos) using a SpeedVac (SOP0324) with high temperature.

2.9.7. Resuspend in:

Component	Amount
2X Hybridization buffer (vial 5)	7.5µl
Hybridization Component A (vial 6)	3µl

2.9.8. Vortex for 10 seconds then centrifuge.

2.9.9. Place the tube in the +95°C heat block for 10 minutes to denature the DNA.

2.9.10. Centrifuge and transfer the content of the tube to a 0.2ml PCR tube.

2.9.11. Add 4 µl of the xGen Lockdown Probes and top up with water to a final volume of 15µl

2.9.12. Mix by pipetting.

2.9.13. The tube should contain the following:

Component	Amount
Multiplex DNA Sample Library Pool	500ng*
Cot-1 DNA	5µg*
xGen® Universal Blockers - TS Mix	2µl*
2X Hybridization Buffer (vial 5)	7.5µl
Hybridization Component A (vial 6)	3µl
xGen Lockdown Probes	4µl
Nuclease-Free Water	0.5µl
Total	**15µl**

**Dried in the SpeedVac*

2.9.14. Incubate hybridization reaction at 47°C in a thermocycler (lid heated at 57°C) for 4 hours (or overnight)

2.9.15. **If proceeding after a 4h hybridization,** change the heat block's temperature used in step 2.9.8 to +47°C.

2.9.16. Allow time to equilibrate to the set temperature.

2.10. Prepare wash buffers

2.10.1. **If proceeding after an overnight hybridization,** turn on a heat block that takes 1.5ml tubes to +47°C, and let it equilibrate to the set temperature.

2.10.2. 2h before the end of the hybridization, dilute 10X Wash Buffers (I, II, III and Stringent) and 2.5X Bead Wash Buffer (Roche, SeqCap EZ Hybridization and Wash Kit, stored at -20^{o}C) to create 1X working solutions.

	Single Capture (+10%)			Two Captures (+10%)		
	Buffer (µl)	Water (µl)	1X Buffer (µl)	Buffer (µl)	Water (µl)	1X Buffer (µl)
10X Wash Buffer I (vial 1)	33	297	330	66	594	660
10X Wash Buffer II (vial 2)	22	198	220	44	396	440
10X Wash Buffer III (vial 3)	22	198	220	44	396	440
10X Stringent Wash Buffer (vial 4)	44	396	440	88	792	880
2.5X Bead Wash Buffer (vial 7)	220	330	550	440	660	1100

2.10.3. For each capture reaction, preheat the following wash buffers to 47°C:

- 1X Stringent Wash Buffer (all)
- 110ul of 1X Wash Buffer I

2.10.4. Equilibrate buffers at 47°C for at least 2 hours before starting wash steps of the captured DNA (step 2.13).

2.11. Prepare the Streptavidin Dynabeads

2.11.1. Allow Dynabeads M-270 Streptavidin (Life Technologies, stored at 4°C) to equilibrate to room temperature for 30 minutes before use (~30minutes before the end of the hybridization)

2.11.2. Mix the beads thoroughly by vortexing for 15 sec.

2.11.3. Aliquot 100µl streptavidin beads per capture into a single 1.5 ml tube (i.e., for 1 capture use 100µl beads, for 2 captures use 200µl beads, etc.).

2.11.4. Place the tube in a magnetic separation rack. Allow the beads to separate from the supernatant. Carefully remove and discard the clear supernatant ensuring that all of the beads remain in the tube.

2.11.5. Add 200µl 1X Bead Wash Buffer per 100µl beads. Vortex for 10 seconds.

2.11.6. Place the tube back in the magnetic rack to bind the beads. Allow the beads to separate from the supernatant. Carefully remove and discard the clear supernatant ensuring that all of the beads remain in the tube.

2.11.7. Repeat Steps 2.11.5 - 2.11.6

2.11.8. After removing the buffer following the second wash, add 1X the original volume of beads of 1X Bead Wash Buffer (i.e., for 100µl beads, use 100µl buffer) and resuspend by vortexing.

2.11.9. Transfer 100µl of the resuspended beads into a new 0.2 ml tube for each capture reaction.

2.11.10. Place the tube in a magnetic rack to bind the beads. Allow the beads to separate from the supernatant. Carefully remove and discard the clear supernatant ensuring that all of the beads remain in the tube.

2.12. <u>Bind hybridized target to the streptavidin beads</u>

2.12.1. Transfer the hybridization samples from step 2.9.14 to the tube containing prepared streptavidin beads.

2.12.2. Mix thoroughly by pipetting up and down 10 times.

2.12.3. Place the tube into a thermal cycler set to 47°C for 45 min (set heated lid at 57°C) to bind the DNA to the beads.

2.12.4. Vortex the tube for 3 seconds every 15 min to ensure that the beads remain in suspension.

2.13. <u>Wash streptavidin beads to remove unbound DNA</u>

2.13.1. Add 100µl pre-heated 1X Wash Buffer I to the tube and vortex for 10 sec to mix.

2.13.2. Transfer the mixture to a fresh low-bind 1.5ml tube.

2.13.3. Place the tube in the magnetic separation rack. Allow the beads to separate from the supernatant. Using a pipette, remove the supernatant containing unbound DNA and discard.

2.13.4. Perform the following wash:

2.13.5. Add 200µl preheated 1X Stringent Wash Buffer and pipette up and down 10 times to mix. Incubate at 47°C for 5 min.

2.13.6. Place the tube in the magnetic separation rack. Allow the beads to separate from the supernatant. Using a pipette, remove the supernatant containing unbound DNA and discard.

2.13.7. Repeat step 2.13.5-2.13.6

2.13.8. Add 200µl room temperature 1X Wash Buffer I and vortex for 2 min to mix.

2.13.9. Place the tube in the magnetic separation rack. Allow the beads to separate from the supernatant. Using a pipette, remove the supernatant and discard.

2.13.10. Add 200µl room temperature 1X Wash Buffer II and vortex for 1 min to mix.

2.13.11. Place the tube in the magnetic separation rack. Allow the beads to separate from the supernatant. Using a pipette, remove the supernatant and discard.

2.13.12. Add 200µl room temperature 1X Wash Buffer III and vortex for 30 sec to mix.

2.13.13. Place the tube in the magnetic separation rack. Allow the beads to separate from the supernatant. Using a pipette, remove the supernatant and discard.

2.13.14. Remove the tube from the magnetic rack and add 25µl Nuclease-Free Water to resuspend the beads. Mix thoroughly by pipetting up and down 10 times.

2.13.15. **Do not pellet or remove the beads**: the post-capture PCR is done on the beads

SAFE STOPPING POINT! 4°C OVERNIGHT OR STORE AT -20°C

2.14. Post-capture PCR

2.14.1. Thaw the KAPA HiFi HotStart ReadyMix and the Post-LM-PCR Oligos from the SeqCap EZ Accessory Kits v2 (stored at -20°C)

2.14.2. Set up the PCR as follow:

	Component	Volume
EACH	Captured library on beads	**25**
MASTER MIX	KAPA HiFi HotStart ReadyMix (Roche)	50
	Post-LM-PCR Oligos 1&2, 5µM (Roche)	5
	Nuclease free water	20
	Total	75 +25 library =100µl / reaction

2.14.3. PCR incubation:

Step	Temperature	Time	No. cycles
Initial Denaturation	98 °C	45 seconds	1
Denaturation	*98 °C*	*15 seconds*	*12*
Annealing	*60 °C*	*30 seconds*	
Extension	*72 °C*	*30 seconds*	
Final Extension	72 °C	1 minutes	1
Hold	10 °C	∞	

SAFE STOPPING POINT! STORE AT -20°C or 4°C

2.14.4. Perform Ampure XP clean-up with 0.68x ratio (refer to NOTE to 2.8.3 about ratio of beads in relation to library quality) and elute into 20µl.

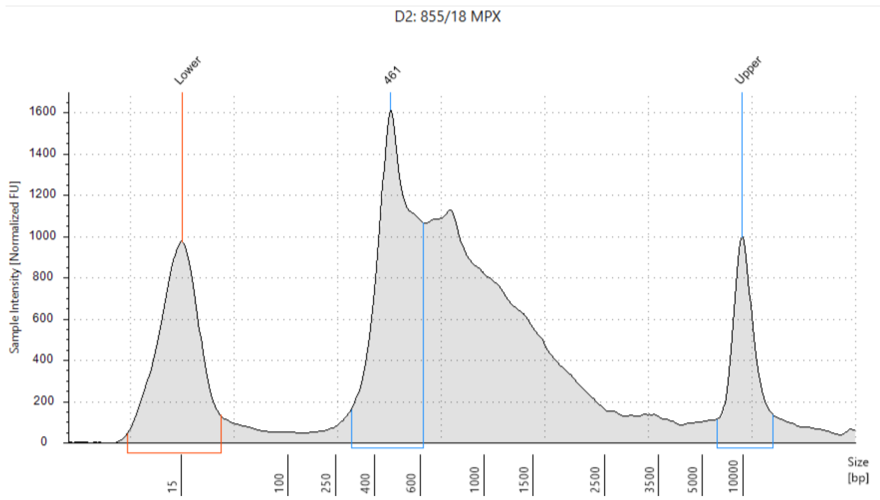
SAFE STOPPING POINT! STORE AT -20°C or 4°C

2.15. Captured Pool QC and 10nM-ing

2.15.1. Quality check the captured pool using Qubit and Tapestation.

2.15.2. Pools concentrations is highly variable and dependent on samples viral load (< 1ng/µl is **NOT** to be considered a fail!)

2.15.3. Tapestation profile should have a long tail, but it's dependent of the quality of the starting RNA. For high quality RNA, after two 0.68x Ampure XP cleanups, one pre- and one post-capture, the profile should look like this:



3. References

3.1. Supplementary Documents

- Library Preparation – Takara-Clontech, SMARTer Stranded Total RNA-Seq Kit v2 – Pico Input Mammalian
- Roche, SeqCap EZ Library SR